

MATH 281 Applied Boundary Value Problems

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1 Heat Equation

Definition 1.1. A **partial differential equation** (PDE) is an equation containing partial derivatives.

1.1 Conduction of Heat in a 1-D Rod

Definition 1.2. The **thermal energy density** $e(x, t)$ is the amount of thermal energy per unit volume.

If we have 1 dimensional rod with cross sectional area A and length L and a slice on the rod from x to $x + \Delta x$, then the heat energy of this slice is:

$$\text{heat energy} = e(x, t)A\Delta x$$

If there is no lateral heat flow on the rod, then the rate of change of the heat energy of this slice must be the sum of the next heat flux into the slice and the heat generate in the slice:

$$\frac{\partial}{\partial t}[e(x, t)A\Delta x] \approx \Phi(x, t)A - \Phi(x + \Delta x, t)A + Q(x, t)A\Delta x$$

Where $Q(x, t)$ is the heat energy generated per unit volume.

We can easily simplify this by dividing by $A\Delta x$ and taking the $\lim_{\Delta x \rightarrow 0}$ to get the following PDE:

$$\frac{\partial e}{\partial t} = -\frac{\partial \Phi}{\partial x} + Q$$

We can also derive this equation by considering a finite slice from $x = a$ to $x = b$. The rate of change of heat energy of this slice is:

$$\frac{d}{dt} \int_a^b e dx = \Phi(a, t) - \Phi(b, t) + \int_a^b Q dx$$

This is also known as the **integral conservation law**.

We can convert this to a single integral by noting that $\frac{d}{dt} \int_a^b e(x, t) = \int_a^b \frac{\partial e(x, t)}{\partial t}$ and that $\Phi(a, t) - \Phi(b, t) = -\int_a^b \frac{\partial \Phi}{\partial x} dx$, resulting in:

$$\int_a^b \left(\frac{\partial e}{\partial t} + \frac{\partial \Phi}{\partial x} - Q \right) dx = 0$$

Since this must be true for any arbitrary a and b , the integrand must be zero:

$$\frac{\partial e}{\partial t} = -\frac{\partial \Phi}{\partial x} + Q$$

Because we also describe the temperatures of materials instead of heat energy, it is useful to convert our equations. We let:

$$\begin{aligned} u(x, t) &\equiv \text{temperature} \\ c &\equiv \text{specific heat capacity} \\ \rho &\equiv \text{density} \end{aligned}$$

Then, the thermal energy density can be expressed as:

$$e(x, t) = c(x)\rho(x)u(x, t)$$

Therefore, the conservation of thermal energy can be expressed as:

$$c(x)\rho(x)\frac{\partial u}{\partial t} = -\frac{\partial \Phi}{\partial x} + Q$$

Theorem 1.1. Fourier's law of heat conduction

$$\Phi = -K_0 \frac{\partial u}{\partial x}$$

Where K_0 is the thermal conductivity of the material.

Substituting this in, we get:

$$c\rho \frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(K_0 \frac{\partial u}{\partial x} \right) + Q$$

If c , ρ , and K_0 are all constant, then this simplifies to the heat equation:

$$\frac{\partial u}{\partial t} = k \frac{\partial^2}{\partial x^2}$$

where

$$k = \frac{K_0}{c\rho}$$

is the thermal diffusivity.